CEFR Test → Score → Report Workflow

Purpose: End-to-end flow from test delivery on platform to final candidate reports (PDF).
Two separate reports: Test 1 (LRW) report and Test 2 (Speaking) report. No combined/overall CEFR.
Reporting layer: Handled outside the engine — Power BI, Claude artifacts, or Excel export.

Phase 1 — Test Delivery (Platform)

Candidate takes two separate tests. All questions pre-loaded from the question bank.

Test 1: Listening + Reading + Writing (LRW)

Section 1: Writing

QB	Questions
Email	1
Essay	1

→ Upstream AI engine outputs section-level `cefr_level` + `raw_score` (0–100).
⚠ **Open item:** Need data from engineering on how section-level score/percentage and CEFR assignment are calculated upstream.

Section 2: Reading (24 questions across 6 passages, 26 marks)

Level	Passage	Parameter	Qs	Type
A2	Passage 1	Topic identification	1	MCQ
	Passage 1	Specific information	1	MCQ
	Passage 1	Gist	1	MCQ
	Passage 2	Main idea	1	MCQ
	Passage 2	Specific information	1	MCQ
	Passage 2	Gist	1	MCQ
B1	Passage 1	Gist	1	MCQ
	Passage 1	Specific information	1	MCQ
	Passage 1	Specific information	1	MCQ
	Passage 1	Detail	1	MCQ
	Passage 2	Gist	1	MCQ
	Passage 2	Specific information	1	MCQ
	Passage 2	Cause and effect	1	MCQ
	Passage 2	Detail	1	MCQ
	Passage 2	Writer's message	1	MCQ
B2	Passage 1	Gist	1	MCQ
	Passage 1	Specific information	1	MCQ
	Passage 1	Inference	1	MCQ
	Passage 1	Writer's stance	1	MCQ
	Passage 2	Gist	1	MCQ
	Passage 2	Specific information	1	MCQ
	Passage 2	Contrasting views	1	**MAQ**
	Passage 2	Inference	1	**MAQ**
	Passage 2	Writer's conclusion	1	MCQ

Totals: A2 = 6 MCQ (6 marks) · B1 = 9 MCQ (9 marks) · B2 = 7 MCQ + 2 MAQ (11 marks)
MAQ: 5 options, 2 correct, partial credit (0/1/2 marks each). Rendered as checkboxes.

Section 3: Listening (22 questions across 8 audios, 26 marks)

Level	Audio	Parameter	Qs	Type
A2	Audio 1	Specific information	1	MCQ
	Audio 1	Likes/dislikes/preference	1	MCQ
	Audio 1	Main topic / gist	1	MCQ
	Audio 2	Speaker's purpose	1	MCQ
	Audio 2	Specific information	1	MCQ
	Audio 2	Understanding direct instructions	1	MCQ
B1	Audio 1	Speaker agreement	1	MCQ
	Audio 1	Specific information	1	MCQ
	Audio 1	Problem identified	1	MCQ
	Audio 2	Gist / main purpose	1	MCQ
	Audio 2	Technical/operational info	1	MCQ
	Audio 2	Decision reached	1	MCQ
	Audio 3	Gist / main issue	1	MCQ
	Audio 3	Technical/operational info	1	MCQ
	Audio 3	Action decided	1	MCQ
B2	Audio 1	Speaker's attitude	1	MCQ
	Audio 1	Specific information (multiple)	1	**MAQ**
	Audio 1	Inference/implication	1	MCQ
	Audio 2	Speaker's tone	1	MCQ
	Audio 2	Specific information (multiple)	1	**MAQ**
	Audio 2	Speaker's evaluation	1	MCQ
	Audio 3	Specific information (multiple)	1	MCQ
	Audio 3	Following detailed instructions	1	MCQ
	Audio 3	Sequence / chronological order	1	MCQ

Totals: A2 = 6 MCQ (6 marks) · B1 = 9 MCQ (9 marks) · B2 = 7 MCQ + 2 MAQ (11 marks)
MAQ: 5 options, 2 correct, partial credit (0/1/2 marks each). Rendered as checkboxes.

Test 2: Speaking (Separate Test)

→ Upstream AI engine outputs section-level `cefr_level` + `raw_score` (0–100).
⚠ **Open item:** Same as Writing — need clarity from engineering on upstream scoring logic.

Phase 2A — Listening & Reading: Consecutive Pass Chain

Same logic for both skills. Runs independently per skill, server-side, on full section submission.

Step 1: Mark each level.
Unanswered = incorrect. Calculate marks per level.

Step 2: Run consecutive pass chain.

For each level in [A1, A2, B1, B2, C1, C2]:
→ if status == skip_pass → PASS, continue
→ if marks >= threshold → PASS, continue
→ if marks < threshold → FAIL, STOP
→ if status == skip_fail → FAIL, STOP

final_level = last level that passed

Thresholds:
A1 → skip_pass
A2 → ⩾ 5/6 (75%)
B1 → ⩾ 6/9 (65%)
B2 → ⩾ 7/11 (63%)
C1 → skip_fail
C2 → skip_fail

There are exactly **5 possible outcomes** per skill (Listening and Reading are identical in structure):

Case 1 — A2 fails → Below A2

A1: auto-pass ✓
A2: 0–4/6 (fail) ✗ → STOP
B1: ignored
B2: ignored
→ display = Below A2, scale band 100–119
→ numerator = A2 marks only, denominator = 6
→ scale range: 100 (0/6) → 113 (4/6)

Case 2 — A2 passes, B1 fails → A2

A1: auto-pass ✓
A2: 5–6/6 (pass) ✓
B1: 0–5/9 (fail) ✗ → STOP
B2: ignored
→ display = A2, scale band 120–139
→ numerator = A2 + B1 marks, denominator = 15
→ scale range: 126 (5+0) → 134 (6+5)

Case 3 — A2 passes, B1 passes, B2 fails → B1

A1: auto-pass ✓
A2: 5–6/6 (pass) ✓
B1: 6–9/9 (pass) ✓
B2: 0–6/11 (fail) ✗ → STOP
→ display = B1, scale band 140–159
→ numerator = A2 + B1 + B2 marks, denominator = 26
→ scale range: 148 (5+6+0) → 155 (6+9+6)

Case 4 — All pass, B2 marks 7–8/11 → B2

A1: auto-pass ✓
A2: 5–6/6 (pass) ✓
B1: 6–9/9 (pass) ✓
B2: 7–8/11 (pass) ✓
C1: auto-fail ✗ → STOP
→ display = B2, scale band 160–169
→ numerator = A2 + B1 + B2 marks, denominator = 26
→ scale range: 166 (5+6+7) → 168 (6+9+8)

Case 5 — All pass, B2 marks 9–11/11 → B2+

A1: auto-pass ✓
A2: 5–6/6 (pass) ✓
B1: 6–9/9 (pass) ✓
B2: 9–11/11 (pass) ✓
C1: auto-fail ✗ → STOP
→ display = B2+, scale band 170–179
→ numerator = A2 + B1 + B2 marks, denominator = 26
→ scale range: 177 (5+6+9) → 179 (6+9+11)

No other outcome is possible. A1 always passes (skip_pass), C1/C2 always fail (skip_fail).

Edge cases for QA:

- All unanswered → all marked wrong → A2 = 0/6 → Case 1 → scale = 100 (floor)
- Perfect score → 6+9+11 = 26/26 → Case 5 → scale = 179 (ceiling)
- Exact threshold at each level (5/6, 6/9, 7/11) → passes, since rule is ⩾

- B2 exactly 9/11 → crosses into B2+ display
- B2 exactly 8/11 → stays B2 display
- Passing B2 with failed A2 or B1 → impossible, chain stops at failure
- Scoring Pre A1, C1, or C2 on L/R → impossible, chain can't reach those

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Phase 2B — Writing & Speaking: Cambridge Scale Mapping Only

Engine receives from upstream AI: `{ cefr_level, raw_score }`
⚠ How upstream derives these values is an open item — pending engineering input.

Writing/Speaking always display B2 (never B2+). B2+ is Listening/Reading only.

Phase 2C — Scale Score Formula (All 4 Skills)

Each skill uses the same base formula but with different inputs.

Listening

Formula:
performance% = total_marks_scored ÷ denominator × 100
scale_score = band_start + ROUND(performance% × band_range ÷ 100)

Display	Denominator	band_start	band_range
Below A2	6	100	19
A2	15	120	19
B1	26	140	19
B2	26	160	9
B2+	26	170	9

Example — Listening B2+ (A2: 6/6, B1: 9/9, B2: 10/11):
total_marks = 6 + 9 + 10 = 25
performance% = 25 ÷ 26 × 100 = 96.2%
scale_score = 170 + ROUND(96.2 × 9 ÷ 100) = 170 + 9 = 179
→ Display: B2+, Scale: 179

Example — Listening B1 (A2: 6/6, B1: 6/9, B2: 4/11 fail):
total_marks = 6 + 6 + 4 = 16
performance% = 16 ÷ 26 × 100 = 61.5%

scale_score = 140 + ROUND(61.5 × 19 ÷ 100) = 140 + 12 = 152
→ Display: B1, Scale: 152

Example — Listening Below A2 (A2: 3/6 fail):
total_marks = 3
performance% = 3 ÷ 6 × 100 = 50.0%
scale_score = 100 + ROUND(50.0 × 19 ÷ 100) = 100 + 10 = 110
→ Display: Below A2, Scale: 110

Reading

Formula:
performance% = total_marks_scored ÷ denominator × 100
scale_score = band_start + ROUND(performance% × band_range ÷ 100)

Display	Denominator	band_start	band_range
Below A2	6	100	19
A2	15	120	19
B1	26	140	19
B2	26	160	9
B2+	26	170	9

Example — Reading B2 (A2: 6/6, B1: 9/9, B2: 7/11):
total_marks = 6 + 9 + 7 = 22
performance% = 22 ÷ 26 × 100 = 84.6%
scale_score = 160 + ROUND(84.6 × 9 ÷ 100) = 160 + 8 = 168
→ Display: B2, Scale: 168

Example — Reading A2 (A2: 5/6, B1: 3/9 fail):
total_marks = 5 + 3 = 8
performance% = 8 ÷ 15 × 100 = 53.3%
scale_score = 120 + ROUND(53.3 × 19 ÷ 100) = 120 + 10 = 130
→ Display: A2, Scale: 130

Example — Reading B2+ (A2: 6/6, B1: 9/9, B2: 10/11):
total_marks = 6 + 9 + 10 = 25
performance% = 25 ÷ 26 × 100 = 96.2%
scale_score = 170 + ROUND(96.2 × 9 ÷ 100) = 170 + 9 = 179
→ Display: B2+, Scale: 179

Writing

Formula:
performance% = (raw_score − band_min) ÷ (band_max − band_min) × 100
scale_score = band_start + ROUND(performance% × band_range ÷ 100)

CEFR	band_min	band_max	band_start	band_range
C2	91	100	200	30
C1	76	90	180	19
B2	56	75	160	19
B1	47	55	140	19
A2	35	46	120	19
A1	20	34	100	19
Pre A1	0	19	10	89

Example — Writing B2 (raw_score: 68, cefr_level: B2):
band_min = 56, band_max = 75
performance% = (68 − 56) ÷ (75 − 56) × 100 = 12 ÷ 19 × 100 = 63.2%
scale_score = 160 + ROUND(63.2 × 19 ÷ 100) = 160 + 12 = 172
→ Display: B2, Scale: 172

Example — Writing A1 (raw_score: 25, cefr_level: A1):
band_min = 20, band_max = 34
performance% = (25 − 20) ÷ (34 − 20) × 100 = 5 ÷ 14 × 100 = 35.7%
scale_score = 100 + ROUND(35.7 × 19 ÷ 100) = 100 + 7 = 107
→ Display: A1, Scale: 107

Example — Writing C1 (raw_score: 82, cefr_level: C1):
band_min = 76, band_max = 90
performance% = (82 − 76) ÷ (90 − 76) × 100 = 6 ÷ 14 × 100 = 42.9%
scale_score = 180 + ROUND(42.9 × 19 ÷ 100) = 180 + 8 = 188
→ Display: C1, Scale: 188

Speaking

Formula:
performance% = (raw_score − band_min) ÷ (band_max − band_min) × 100
scale_score = band_start + ROUND(performance% × band_range ÷ 100)

CEFR	band_min	band_max	band_start	band_range
C2	80	100	200	30
C1	70	79	180	19
B2	59	69	160	19
B1	46	58	140	19
A2	31	45	120	19
A1	20	30	100	19
Pre A1	0	19	10	89

Example — Speaking B1 (raw_score: 52, cefr_level: B1):
band_min = 46, band_max = 58
performance% = (52 − 46) ÷ (58 − 46) × 100 = 6 ÷ 12 × 100 = 50.0%
scale_score = 140 + ROUND(50.0 × 19 ÷ 100) = 140 + 10 = 150
→ Display: B1, Scale: 150

Example — Speaking A2 (raw_score: 38, cefr_level: A2):
band_min = 31, band_max = 45
performance% = (38 − 31) ÷ (45 − 31) × 100 = 7 ÷ 14 × 100 = 50.0%
scale_score = 120 + ROUND(50.0 × 19 ÷ 100) = 120 + 10 = 130
→ Display: A2, Scale: 130

Example — Speaking B2 (raw_score: 65, cefr_level: B2):
band_min = 59, band_max = 69
performance% = (65 − 59) ÷ (69 − 59) × 100 = 6 ÷ 10 × 100 = 60.0%
scale_score = 160 + ROUND(60.0 × 19 ÷ 100) = 160 + 11 = 171
→ Display: B2, Scale: 171

⚠ Writing and Speaking always display B2 (never B2+). B2+ is Listening/Reading only.

Phase 3 — Report Generation (Outside Engine)

Two separate reports generated.

Report 1: Test 1 — LRW Report

Page 1 — Cover

- iMocha logo
- Product name: "AI-English Pro"

- Candidate Name, Candidate ID, Company, Date of Attempt

Page 2 — Score Analysis

- Bar chart: CEFR level per skill (Writing, Reading, Listening)
- Y-axis: Pre A1 → C2
- CEFR scale legend strip (Pre A1 through C2 with labels)
- One-liner: CEFR definition

Page 3 — Section Skill Analysis (Writing, Reading)

For each skill, show:

- Section number + skill name
- Total Score: `raw_score`/100 (W) or derived percentage (R)
- CEFR badge: `cefr_display` + proficiency label
- Candidate capability statement (templated per CEFR level per skill)
- Skill definition sidebar

Page 4 — Section Skill Analysis (Listening) + Proctoring + Test Log

- Listening section (same format as above)
- Proctoring Analysis: images captured, image violations, window/time violations
- Test Log: appeared on, completed on, report generated on, IP address

Report 2: Test 2 — Speaking Report

Page 1 — Cover

- Same layout as Report 1
- Product name: "AI-English Pro"
- Candidate Name, Candidate ID, Company, Date of Attempt

Page 2 — Score Analysis

- CEFR level for Speaking
- CEFR scale legend strip

Page 3 — Section Skill Analysis (Speaking)

- Total Score: `raw_score`/100
- CEFR badge: `cefr_display` + proficiency label

- Candidate capability statement
- Skill definition sidebar
- Proctoring Analysis + Test Log

Proficiency Labels (Both Reports)

CEFR	Label
Pre A1 / A1	Beginner User
A2	Elementary User
B1	Intermediate User
B2 / B2+	Independent User
C1	Proficient User
C2	Mastery

Candidate Capability Statements

Template bank needed per skill × per CEFR level. *(To be authored — 7 levels × 4 skills = 28 statements.)*